

Graph Neural Networks for Molecular Property Prediction

Abstract

We present a deep learning approach using Graph Convolutional Networks (GCN) to predict molecular toxicity with high accuracy.

Introduction

Predicting molecular properties early in the drug discovery pipeline reduces costs and accelerates development. Traditional methods rely on expensive experiments or simple molecular descriptors.

Methodology

Molecules are represented as undirected graphs $G=(V,E)$ where nodes V represent atoms and edges E represent chemical bonds.

Results

Our model achieves AUC-ROC of 0.94 on Tox21 and 0.91 on BBBP. Ablation studies confirm attention-based readout contributes significantly to performance.

Conclusion

Graph neural networks are powerful tools for molecular property prediction. The proposed architecture generalizes across diverse chemical spaces.

References

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